Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Rosenmund reduction is used for the preparation of

(A) carboxylic acid

(B) aldehydes

(C) esters

(D) alcohols

Q.2

Reduction of acetyl chloride with H_2 in presence of Pd and $BaSO_4$ gives:

(A) CH_3COCH_3

(B) C_2H_5OH

(C) CH_2COOH

(D) CH_3CHO

Q.3

Catalyst $SnCl_2/HCl$ is used in which of the following method of synthesis of aldehyde?

(A) Stephen's reduction

(B) Cannizzaro reaction

(C) Clemmensen's reduction

(D) Rosenmund's reduction

Q.4

The incorrect statement regarding oxo process for synthesis of an aldehyde is

(A) Mixture of CO and H_2 is allowed to react with an alkene

(B) $Co_2(CO)_8$ may be used as a catalyst

(C) $[Co(CO)_4H]$ may act as a catalyst

(D) This process can also be used in the same manner for the synthesis of ketone

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Q.5

Which of the following compounds give a ketone with grignard reagent?

(A) Formaldehyde

(B) Ethyl alcohol

(C) Methyl cyanide

(D) Methyl iodide

Q.6

Acetyl chloride reacts with ____ to give butan-2-one.

(A) cadmium chloride

(B) methyl magnesium chloride

(C) dimethyl cadmium

(D) diethyl cadmium

Q.7

Dialkyl cadmium reacts with a compound to form a ketone. The compound is:

(A) an acid

(B) an acid chloride

(C) an ester

(D) carbon monoxide

Q.8

An acetoacetic ester synthesis of a ketone proceeds by alkylation of the enolate of the acetoacetic ester followed by ester hydrolysis and decarboxylation of the β -ketoacid. Which of the following methyl ketones is difficult to prepare by this method?

(A) $CH_3\overset{\overset{O}{||}}{C}CH_2C(CH_3)_3$

(B) $CH_3\overset{\overset{O}{||}}{C}CH_2CH_2Ph$

(C) $CH_3\overset{\overset{O}{||}}{C}CH_2CH_2CH=CH_2$

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(D) $CH_3\overset{O}{\overset{\|}{C}}CH(CH_2Ph)_2$

Q.9

$CH_3 - CH_2C \equiv N \xrightarrow{X} CH_3CH_2CHO$

The compound X is

(A) $SnCl_2/HCl/H_2O$, boil

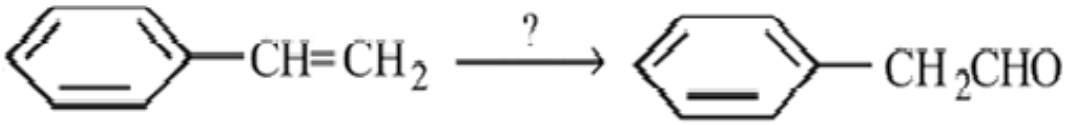
(B) $H_2/Pd - BaSO_4$

(C) $LiAlH_4/$ ether

(D) $NaBH_4/$ ether $/H_3O^+$

Q.10

The correct set of reagents to carry out the following conversion



(A) $B_2H_6/$ diglyme, $H_2O_2/NaOH$, PCC/CH_2Cl_2

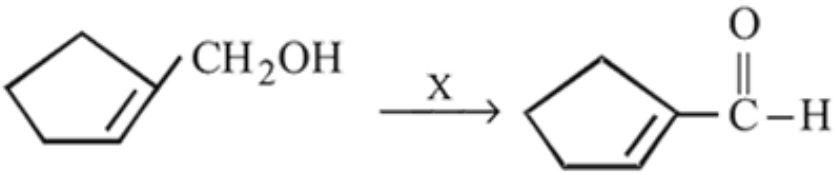
(B) $Br_2, NaNH_2, HgSO_4/H_2SO_4$

(C) $H_2O/H_2SO_4, CrO_3/H_2SO_4$

(D) None of the above

Q.11

What is X in the following conversion?



(A) $K_2Cr_2O_7, H_2SO_4/H_2O$

(B) HIO_4

(C) PCC/CH_2Cl_2

(D) $OsO_4, (CH_3)_3C.COOH, (CH_3)_3COH, OH$

Q.12

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Methyl ethyl ketone is prepared by the oxidation of

(A) 2-propanol

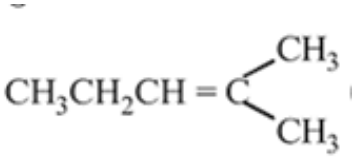
(B) 1 -butanol

(C) 2-butanol

(D) t-butyl alcohol

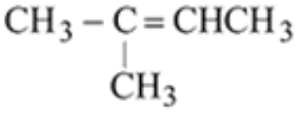
Q.13

Which alkene on ozonolysis gives CH_3CH_2CHO and $CH_3\overset{\overset{O}{\|}}{C}CH_3$?

(A) 

(B) $CH_3CH_2CH=CHCH_2CH_3$

(C) $CH_3CH_2CH=CHCH_3$

(D) 

Q.14

Pinacolone is

(A) 2,3 -Dimethyl-2, 3-butanediol

(B) 3,3-Dimethyl-2-butanone

(C) 1 -Phenyl-2-propanone

(D) 1,1 -Diphenyl-1, 2-ethandiol

Q.15

5 - Methyl - 2 -hexanone can be synthesised from acetoacetic ester and RX. Which of the following RX is used?

(A) $(CH_3)_2CHBr$

(B) $(CH_3)_2CHCH_2Br$

(C) $CH_3CH_2CHBrCH_3$

(D) $(CH_3)_2CHCH_2CH_2Br$

Answers & Solutions

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Q.1 Answer:

aldehydes

Solution:

Rosenmund reduction is used for the preparation of aldehydes. In this reaction, the aldehyde group is obtained by the reduction of an acid chloride using hydrogen in the presence of Lindlar's catalyst.

$$RCOCl + H_2 \xrightarrow{\text{Lindlar scatalyst}} RCHO + HCl$$

Q.2 Answer:

CH_3CHO

Solution:

$$CH_3-\overset{\overset{O}{\|}}{C}-Cl + H_2 \xrightarrow{Pd/BaSO_4} \underset{\text{acetaldehyde}}{CH_3-\overset{\overset{O}{\|}}{C}-H} + HCl$$

This reaction is known as Rosenmund reaction.

Q.3 Answer:

Stephen's reduction

Solution:

Stephen aldehyde synthesis reaction involves the preparation of aldehydes $(R-CHO)$ from nitriles $(R-CN)$ using $tin(II)$ chloride $(SnCl_2)$, hydrochloric acid (HCl) and quenching the resulting iminium salt ($[R\,CH=NH_2]^+\,Cl^-$) with water (H_2O).

Q.4 Answer:

This process can also be used in the same manner
for the synthesis of ketone

Solution:

Oxo process involves formulation of alkene to form aldehyde only not ketone

Q.5 Answer:

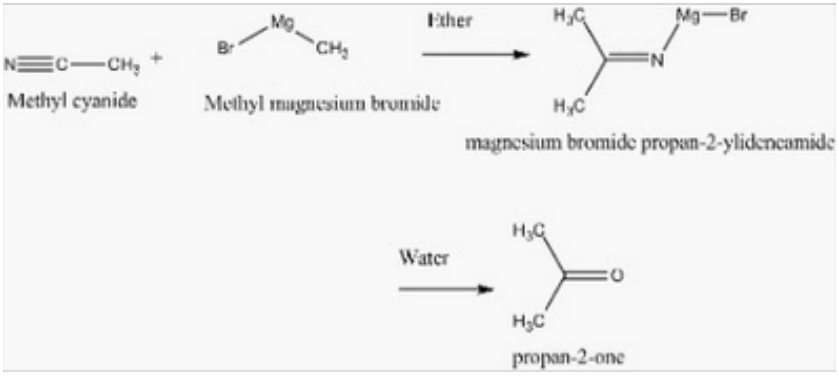
Methyl cyanide

Solution:

Methyl cyanide gives a ketone with grignard reagent. For example, methylcyanide reacts with methyl magnesium bromide to form an adduct magnesium bromide propan-2-ylideneamide which on

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hydrolysis gives propan-2-one.



Q.6 Answer:

diethyl cadmium

Solution:

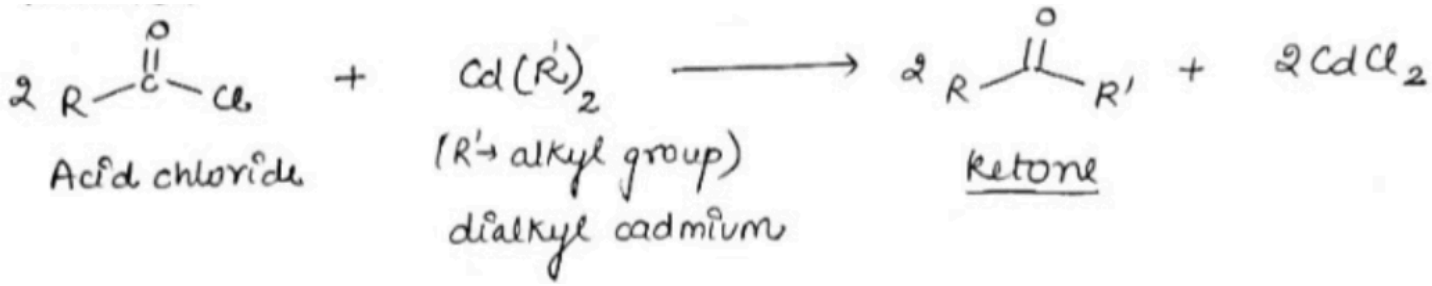
Correct Answer is diethyl cadmium

Q.7 Answer:

an acid chloride

Solution:

Dialkyl cadmium reacts with an acid chloride to form a ketone.



So, the correct option is (B)

Q.8 Answer:

$$CH_3\overset{\overset{O}{\|}}{C}CH(CH_2Ph)_2$$

Solution:

The alkylation of the enolate is an S_N2 reaction, and sterically hindered alkyl groups are difficult to introduce. However, even dialkylation is possible with reactive alkylating agents, as in (d).

Q.9 Answer:

$SnCl_2/HCl/H_2O$, boil

Solution:

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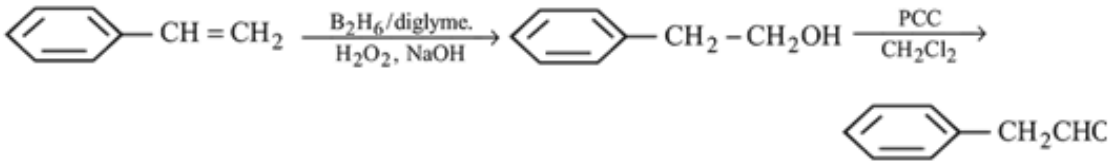
It is Stephen's reaction.

$$CH_3CH_2C \equiv N \xrightarrow{SnCl_2/HCl} CH_3CH_2CH = NH \cdot HCl \xrightarrow{H_2O} CH_3CH_2CHO + NH_4C$$

Q.10 Answer:

B_2H_6/ diglyme, $H_2O_2/NaOH$,
PCC/CH_2Cl_2

Solution:



Q.11 Answer:

PCC/CH_2Cl_2

Solution:

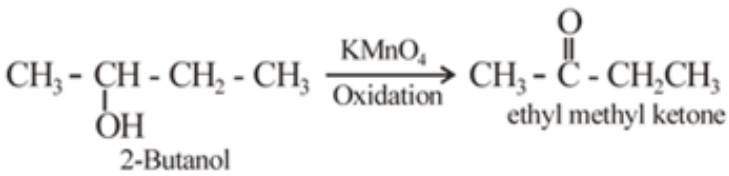
Pyridinium chloro chromate selectively oxidises a primary alcohol to an aldehyde e.g.

$$RCH_2OH \xrightarrow[CH_2Cl_2]{PCC} RCHO$$

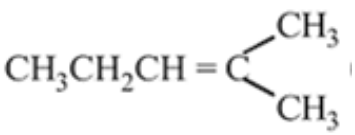
Q.12 Answer:

2-butanol

Solution:

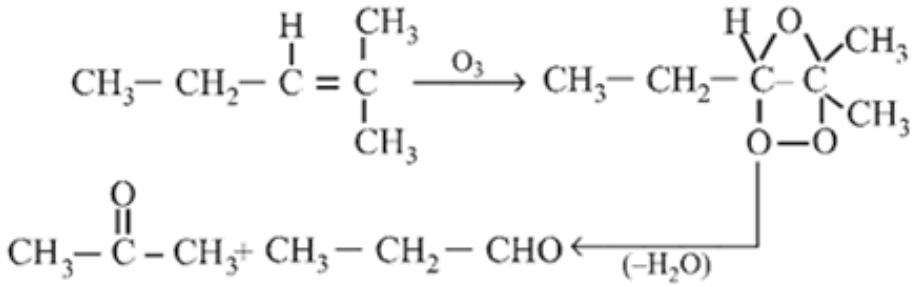


Q.13 Answer:



Solution:

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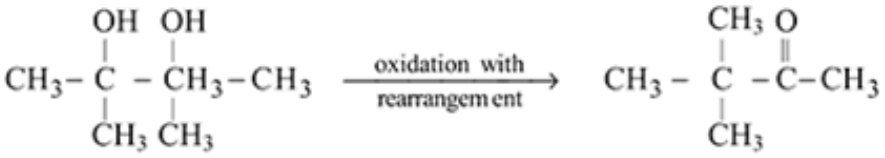


Q.14 Answer:

3,3-Dimethyl-2-butanone

Solution:

Pinacolone is oxidation product of pinacol.



Pinacol

Pinacolone

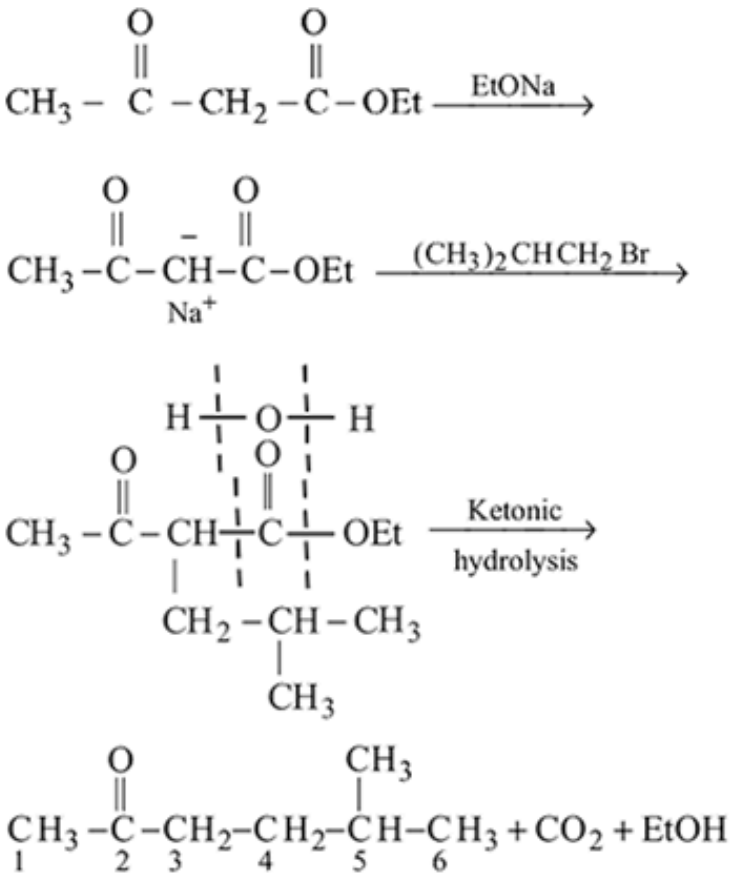
3, 3-dimethyl-2-butanone

Q.15 Answer:

$(CH_3)_2 CHCH_2Br$

Solution:

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